

The Impact of Economic Indicators on Public Health Outcomes: A Global Analysis

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I. Introduction

Economic development and public health are closely intertwined, with economic stability often influencing health outcomes. Countries with strong economies tend to have better healthcare infrastructure, higher public health expenditures, and improved living conditions. On the other hand, economic instability can lead to reduced access to healthcare, malnutrition, and increased disease burden.

This study seeks to explore the relationship between economic indicators (Gross Domestic Product (GDP), unemployment rates, and inflation) and key health indicators (life expectancy, mortality rates, immunization rates, access to clean water, and sanitation) of countries in the world. The central research question is:

How do economic conditions influence health outcomes globally over time?

The hypothesis is that:

1. Countries with higher GDP will have higher life expectancy and lower mortality rates, as economic prosperity enables better healthcare systems.
2. Higher unemployment rates will correlate with poorer health outcomes, as job insecurity often leads to lower income and limited access to healthcare services.
3. Higher inflation rates may negatively impact health indicators by increasing the cost of healthcare and essential goods, reducing affordability for lower-income populations.

The results will be significant as understanding these relationships can help policymakers and international organizations make informed decisions about resource allocation, social welfare policies, and healthcare investments.

II. Methods

Data acquisition

The dataset for this study was collected using the World Bank API. This API provides access to high-quality, globally comparable economic and health statistics. The selected indicators cover a period from 1960 to 2023 for all available countries. The date range was picked in order to include as much observations as possible.

In order to showcase the ability to use different techniques of gathering data, I used the `wbstats` package in R to gather the data for the health indicators, and used `httr` package and the API call method to gather the data for the economic indicators (using url <http://api.worldbank.org/v2>). The source of data for both methods were essentially the same.

The following five health indicators were extracted, with their individual indicator codes provided, as needed for the API call:

- Life Expectancy (SP.DYN.LE00.IN): The average number of years a newborn is expected to live.
- Mortality Rate (SH.DYN.MORT): The number of deaths per 1,000 live births.
- Immunization Rate (SH.IMM.IDPT): The percentage of infants receiving standard immunizations.
- Water Access (SH.H2O.SAFE.ZS): The percentage of the population with access to safe drinking water.
- Sanitation Access (SN.ITK.DEFC.ZS): The percentage of the population with access to improved sanitation facilities.

Additionally, three economic indicators were obtained:

- Gross Domestic Product (GDP) (NY.GDP.MKTP.CD): The total economic output of a country in current USD.
- Unemployment Rate (SL.UEM.TOTL.ZS): The percentage of the labor force that is unemployed.
- Inflation Rate (FP.CPI.TOTL.ZG): The annual percentage change in consumer prices.

Data Cleaning & Wrangling

After data extraction, the following preprocessing steps were applied:

- **Standardization:** Renamed columns for clarity and converted year values to integers to standardize formats across datasets.
- **Merging:** Combined the datasets using `country_code`, `country_name`, and `year` as keys, aligning each indicator into a single wide-format dataset.
- **Handling Missing Data:** We retained missing values (NA) and outliers, as the World Bank is a reliable source. Excluding observations may introduce bias; retaining them allows for analysis of missingness patterns and preserves generality.

The resulting combined dataset, named `merged_data`, included 17280 observations after merging and cleaning.

Data Transformation

Economic indicators—GDP, inflation rate, and unemployment rate—were log-transformed (e.g., `log_gdp`) to reduce skewness and allow for percentage-based interpretation in regression models. Health indicators were left in their original scale for interpretability.

Modeling Approach

We employed two complementary modeling strategies:

1. Multiple Linear Regression (MLR)

For each health outcome, a multiple linear regression model was fit using the predictors `log_gdp`, `log_inflation_rate`, `log_unemployment_rate`, and `year`. The `year` variable, although not being an economic indicator, was added here because it could potentially be an important factor for the health indicators. These models enable interpretability of coefficients, which indicate the direction and strength of association between economic indicators and health outcomes.

- **Model Evaluation Metrics:**
 - R^2 : Proportion of variance explained.
 - RMSE (Root Mean Squared Error): Average squared prediction error.
 - MAE (Mean Absolute Error): Average absolute prediction error.
- **Residual Diagnostics:**

To validate model assumptions, we plotted residuals vs. fitted values and QQ-plots for each linear regression. This allowed us to check for homoscedasticity and normality of residuals. While minor deviations were observed, they did not invalidate the overall modeling strategy.

- **Multicollinearity Check:**
We computed the Variance Inflation Factor (VIF) for all predictors. All VIF values were below 2, indicating that multicollinearity was not a concern.

2. Random Forest Regression

Random forests are ensemble learning methods that use multiple decision trees to capture complex nonlinear relationships and interactions between predictors. We used the default random forest in the `r` package, so no special hyperparameters.

- **Interpretation:** Although less interpretable than linear models, we extracted and visualized variable importance metrics (e.g., `%IncMSE` and `IncNodePurity`) to understand the relative impact of predictors.
- **Evaluation Metrics:** RMSE and MAE on the test set were used for comparison with linear models.

Training and Testing

To ensure model generalizability, we split the dataset into a training set (80%) and a test set (20%) using a stratified sampling approach based on country.

Models were trained on the training set and evaluated on the test set using RMSE and MAE to compare predictive performance across health outcomes.

Tools Used

- **Data Extraction:** `wbstats`, `http`
- **Data Cleaning and Wrangling:** `dplyr`, `tidyr`
- **Modeling:** `lm` for linear regression, `randomForest` for random forest models
- **Model Evaluation:** `yardstick`, `caret`
- **Visualization:** `ggplot2`, `plotly`
- **Table Formatting:** `kableExtra` for presenting results in RMarkdown

Residual analysis revealed some departures from normality, particularly in the Q-Q plots. While this suggests the linear regression assumptions may not be fully satisfied, these models were primarily used for exploratory analysis. The inclusion of random forest models, which do not rely on distributional assumptions, helps support the robustness of our findings.

III. Results

This section presents both exploratory data analysis (EDA) and predictive modeling results. The goal was to understand how economic relate to five key public health indicators: life expectancy, under-5 mortality rate, immunization coverage, access to basic water services, and access to basic sanitation.

Summary of Health Indicators

Table 1: Summary Statistics of Health Indicators

Life Expectancy	Mortality Rate (Under 5)	Immunization Rate (DTP)	Access to Water (%)	Access to Sanitation (%)
Min. :11.99	Min. : 1.50	Min. : 0.00	Min. : 18.68	Min. : 2.794
1st Qu.:57.03	1st Qu.: 17.00	1st Qu.:69.57	1st Qu.: 78.66	1st Qu.: 49.537
Median :66.96	Median : 43.00	Median :87.00	Median : 93.47	Median : 85.375
Mean :64.39	Mean : 71.27	Mean :78.44	Mean : 85.57	Mean : 72.751
3rd Qu.:72.65	3rd Qu.:102.03	3rd Qu.:95.00	3rd Qu.: 98.95	3rd Qu.: 97.437
Max. :85.53	Max. :428.70	Max. :99.00	Max. :100.00	Max. :100.000

Life Expectancy ranges from 11.99 to 85.53 years, with a median of 66.96 years. The distribution appears left-skewed, although the mean (64.39) is slightly lower than the median, suggesting that some countries have significantly lower life expectancy values.

Mortality Rate (Under 5) has a wide spread, ranging from 1.50 to 428.70 deaths per 1,000 live births. The median (43.00) and mean (71.27) suggest a right-skewed distribution, likely influenced by countries with very high child mortality.

Immunization Rate (DTP) is skewed to the left, with most countries having high coverage. The minimum value is 0%, while the maximum is 99%. The median (87%) and third quartile (95%) indicate that many countries achieve high immunization rates.

Access to Water and Sanitation is generally high across countries, with both variables showing median values of 93.47% and 85.38%, respectively. However, the first quartiles (78.66% for water and 49.54% for sanitation) suggest that a significant number of countries still experience lower access, leading to a left-skewed distribution for these variables.

The graphs were omitted here for brevity. Please refer to the Appendix section (Figure 4~8).

Summary of Economic Indicators

Table 2: Summary Statistics of Economic Indicators

GDP (Current US\$)	Unemployment Rate (%)	Inflation Rate (%)
Min. :2.586e+06	Min. : 0.100	Min. : -17.640
1st Qu.:2.294e+09	1st Qu.: 4.083	1st Qu.: 2.409
Median :1.692e+10	Median : 6.341	Median : 4.944
Mean :1.185e+12	Mean : 7.898	Mean : 19.826
3rd Qu.:2.013e+11	3rd Qu.:10.390	3rd Qu.: 9.702
Max. :1.062e+14	Max. :38.800	Max. :23773.132

GDP (Current US\$) shows an extreme right-skew, with values ranging from \$2.6 million to \$106.2 trillion. The median GDP is \$16.9 billion, while the mean is \$1.185 trillion, highlighting the presence of a few very wealthy countries driving up the average. Log transformation could be necessary to normalize the distribution. After normalizing, the distribution appears moderately spread.

Unemployment Rate (%) is moderately spread, ranging from 0.1% to 38.8%, with a median of 6.34%. The distribution is slightly right-skewed, meaning a few countries have significantly higher unemployment rates.

Inflation Rate (%) exhibits high variability, with a minimum of -17.64% (deflation) and a maximum of 23,773% (hyperinflation). The median of 4.94% suggests that most countries experience moderate inflation, but extreme outliers result in a highly right-skewed distribution, further justifying the use of log transformation, after which we get a relatively moderate skewness.

The graphs were omitted here for brevity. Please refer to the Appendix section (Figure 1~3).

Exploratory Analysis of Relationship Between Indicators

Scatterplots and correlation matrices were used to visualize relationships between economic and health indicators. Several consistent patterns emerged:

Life Expectancy showed a strong positive association with log(GDP) and year, and a mild negative relationship with log(Inflation Rate). Countries with higher income levels tended to have longer lifespans. The positive association with year reflects global improvements over time.

Under-5 Mortality Rate was strongly negatively correlated with log(GDP) and year. Countries with lower GDP and earlier years tended to have higher mortality rates. There was also some evidence of positive association with inflation, suggesting that economic instability may reduce child survival.

Immunization Rate, Access to Water, and Access to Sanitation were all positively correlated with GDP and year, though the strength of these associations varied. The scatterplots showed noticeable heterogeneity: some low-GDP countries had high immunization due to donor-funded programs, while others lagged behind. This highlights the potential role of nonlinearities or country-specific factors not captured in simple correlations.

Inflation and unemployment generally showed weak or inconsistent correlations in scatterplots. Inflation tended to have negative associations with most health outcomes, but the relationships were not always linear or monotonic.

The graphs were omitted here for brevity. Please refer to the Appendix section (Figure 9~14).

Regression and Random Forest Modeling

To quantify the effects of the economic indicators on health outcomes, both linear regression and random forest models were fit to the data. All predictors were log-transformed where appropriate, and multicollinearity was checked using Variance Inflation Factor (VIF); all VIFs were below 1.12, indicating no problematic collinearity.

1. Life Expectancy

Adjusted $R^2 = 0.253$

Strong positive effects from log(GDP) (+2.99) and Year (+2.34).

Strong negative impact from log(Inflation) (-2.53), suggesting economic instability reduces lifespan.

Random forest confirmed GDP and Inflation as most important.

2. Under-5 Mortality Rate

Adjusted $R^2 = 0.212$

GDP and Unemployment were strong negative predictors (-9.6 each), meaning that higher income and employment reduce child mortality.

Inflation had a strong positive effect (+9.74), possibly due to reduced access to care or food during inflationary periods.

Year had a strong negative trend (-20.83).

RF results aligned closely, with all predictors contributing significantly.

3. Immunization Rate

Adjusted $R^2 = 0.110$ (lowest fit among the indicators)

GDP and Year had positive effects; Inflation had a strong negative effect (-2.77).

Unemployment showed an unexpected positive effect ($+2.58$), potentially reflecting policy efforts in certain regions.

Scatterplots suggested that some low-income countries achieved high immunization rates due to global vaccine initiatives—highlighting the limits of linear assumptions.

4. Access to Water

Adjusted $R^2 = 0.185$

GDP ($+4.72$) and Unemployment ($+3.70$) had positive effects, while Inflation (-4.97) had a strong negative association.

Year also contributed positively ($+4.93$).

Random forest showed GDP and Inflation as the most important variables, consistent with linear results.

5. Access to Sanitation

Adjusted $R^2 = 0.194$

GDP had the largest positive effect ($+8.28$), followed by Unemployment ($+6.88$) and Year ($+7.39$).

Inflation was again strongly negative (-7.99).

Random forest ranked GDP highest in importance, closely followed by Inflation and Unemployment.

Interpretation

The models reveal distinct and consistent patterns in how economic indicators relate to health outcomes:

GDP is the most reliable and consistently positive predictor. For every 1% increase in GDP, we observe:

A 2.99-year increase in life expectancy.

A 9.59-point reduction in under-5 mortality rate.

A 1.26-point increase in immunization coverage.

A 4.72-point increase in access to basic water services.

A 8.28-point increase in access to basic sanitation.

This confirms that wealthier countries or those with growing economies tend to invest more effectively in public health infrastructure, clean water, sanitation, and child health.

Inflation has a consistently harmful effect on health:

A 1% increase in inflation is associated with a 2.53-year decrease in life expectancy, a 9.74-point increase in child mortality, and significant decreases in water access (−4.97), sanitation (−8.00), and immunization (−2.77).

These results suggest that inflation erodes real income and reduces access to health services, especially in lower-income settings where subsidies and safety nets may be limited.

Unemployment effects vary by outcome:

It increases under-5 mortality by 9.63 points and decreases life expectancy slightly (−0.88), consistent with literature linking joblessness to stress, food insecurity, and lower healthcare access.

However, it has positive coefficients for immunization (+2.58), water (+3.70), and sanitation (+6.88). This could reflect governments or NGOs prioritizing basic service delivery during economic downturns or in countries where informal labor is common and official unemployment doesn't strongly reflect household income.

Year (as a proxy for global progress over time) consistently improves all outcomes:

Each additional year is associated with:

- +2.34 years in life expectancy,
- −20.83 points in child mortality,
- +5.03 points in immunization,
- +4.93 in water access, and
- +7.39 in sanitation.

These trends reflect widespread health gains over the past two decades, likely driven by global development goals, technological advancement, and international aid.

Random forest models confirm GDP and inflation as the most important predictors, especially for mortality and life expectancy, indicating that these variables have both linear and potentially nonlinear impacts across different countries and regions.

IV. Conclusions and Summary

This analysis provides strong empirical evidence that economic conditions—especially GDP, inflation, and unemployment—are closely linked to a wide range of public health outcomes across countries. Among the economic indicators examined, GDP consistently emerged as the strongest and most reliable predictor of better health, including longer life expectancy, lower child mortality, and broader access to immunization, clean water, and sanitation. In contrast, inflation was strongly associated with worse health, particularly increased child mortality and reduced access to essential services. The role of unemployment was more nuanced, showing negative impacts on mortality and life expectancy, but unexpectedly positive associations with infrastructure-related health indicators, suggesting the influence of government response mechanisms or measurement artifacts.

Crucially, these relationships remained consistent across both linear models and random forest algorithms, strengthening confidence in the robustness of the results. Moreover, the year variable consistently showed improvement in health outcomes over time, reflecting long-term progress driven by global development, medical innovation, and public health initiatives.

These findings underscore the interconnectedness of economic policy and public health: improvements in national economic indicators can produce tangible gains in population health, while economic instability—especially inflation—can have far-reaching adverse effects. Policies that stimulate economic growth while curbing inflation may be particularly effective in improving global health outcomes.

Limitations

Despite the strength of the results, several limitations must be acknowledged:

Causality cannot be inferred: These models are correlational and do not account for reverse causality (e.g., better health enabling economic growth).

Unmeasured confounders such as education, political stability, healthcare quality, and income inequality were not included, which may influence both economic indicators and health outcomes.

Country-specific and regional differences were not explicitly modeled. Some associations may vary substantially between low-, middle-, and high-income countries.

Simplified transformations (e.g., log transformation of GDP, inflation, and unemployment) improve model fit but may obscure nonlinear or threshold effects in reality.

Data quality and completeness vary across countries and indicators, which may introduce measurement bias.

Despite these limitations, this project highlights important pathways by which economic conditions shape population health. Future research could incorporate more granular variables (e.g., public health spending, education levels), use country-level fixed effects to better control for unobserved heterogeneity, and explore temporal dynamics through longitudinal or panel modeling approaches.

V. Appendix

Histograms of Variables

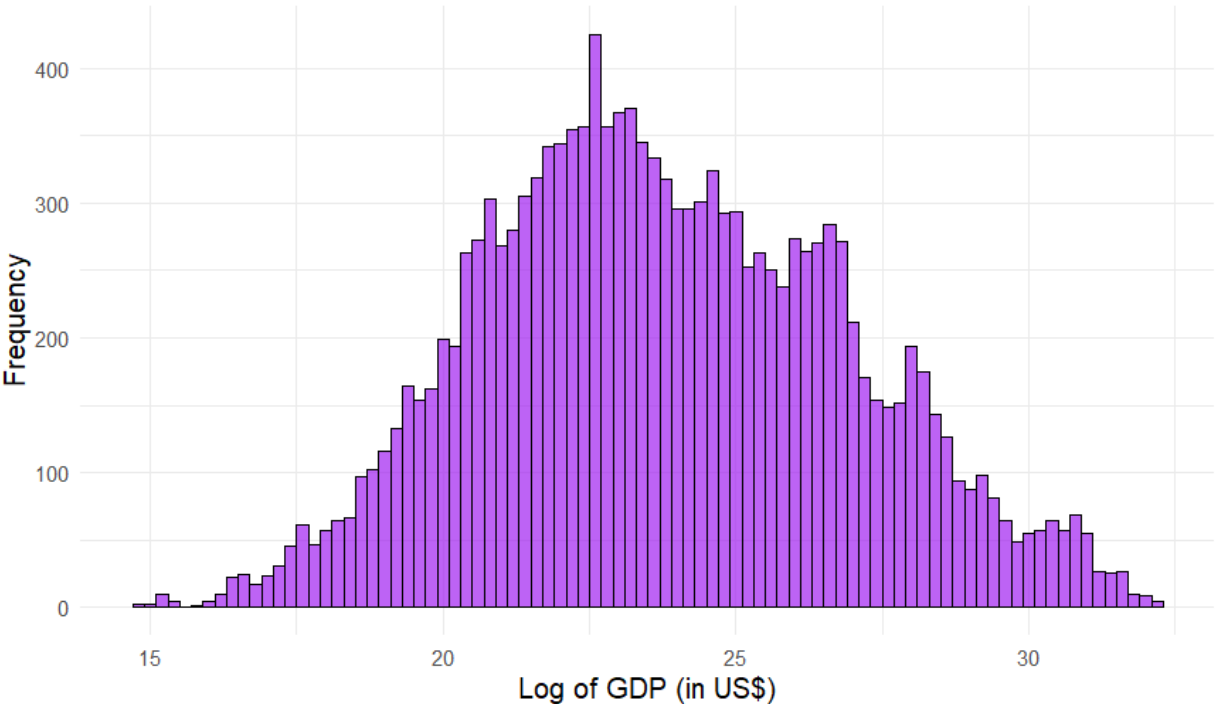


Figure 1: Log-Transformed GDP Distribution (in US\$)

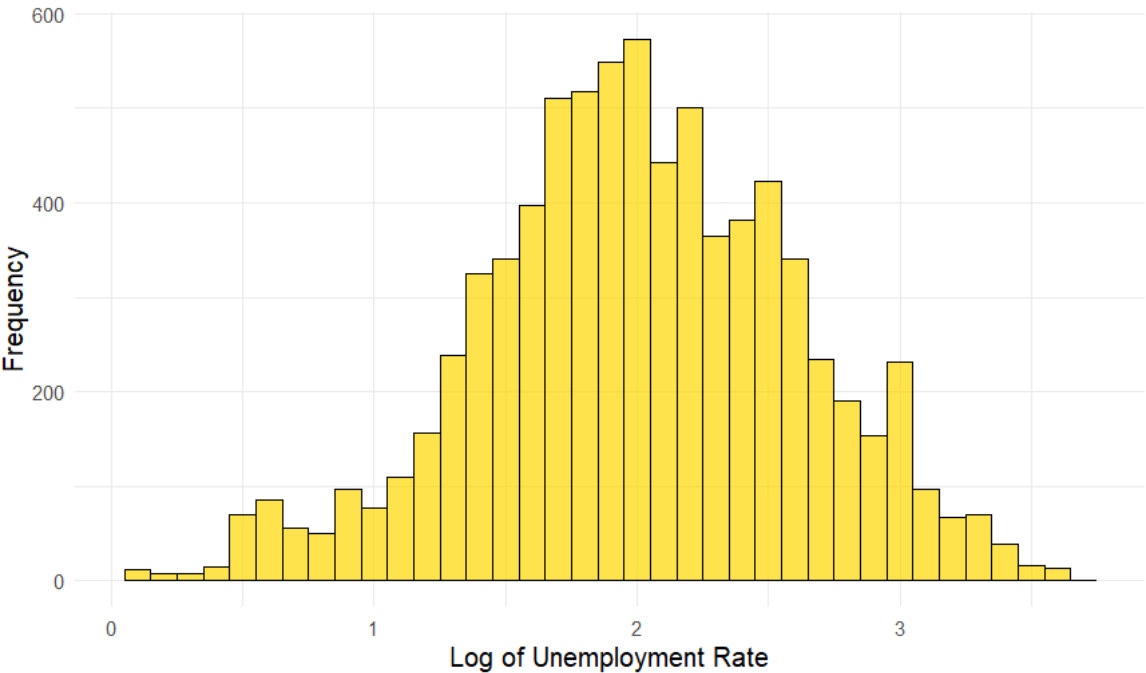


Figure 2: Log-Transformed Unemployment Rate Distribution

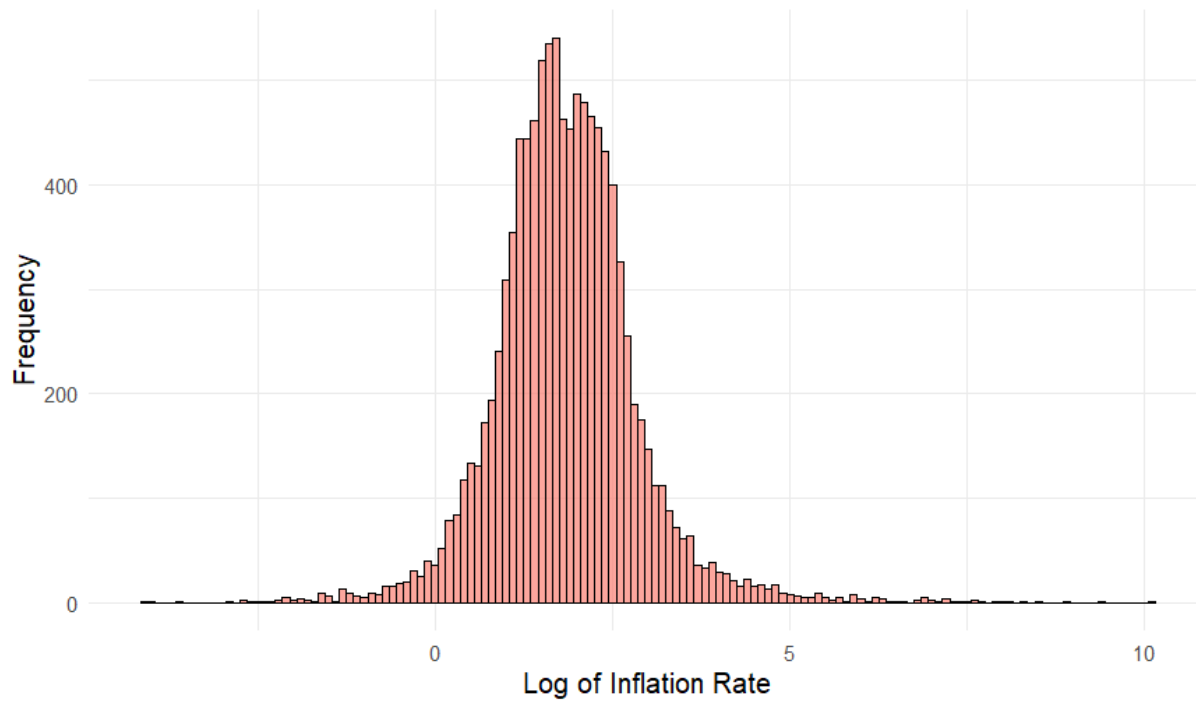


Figure 3: Log-Transformed Inflation Rate Distribution



Figure 4: Life Expectancy Distribution

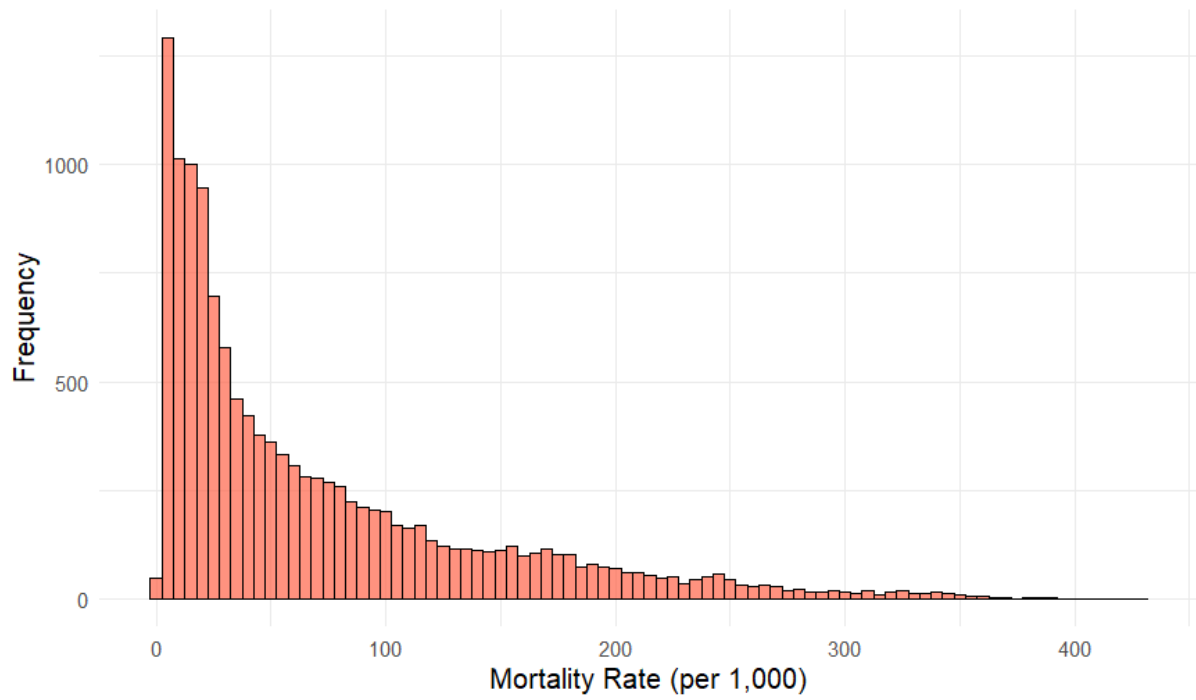


Figure 5: Mortality Rate Distribution

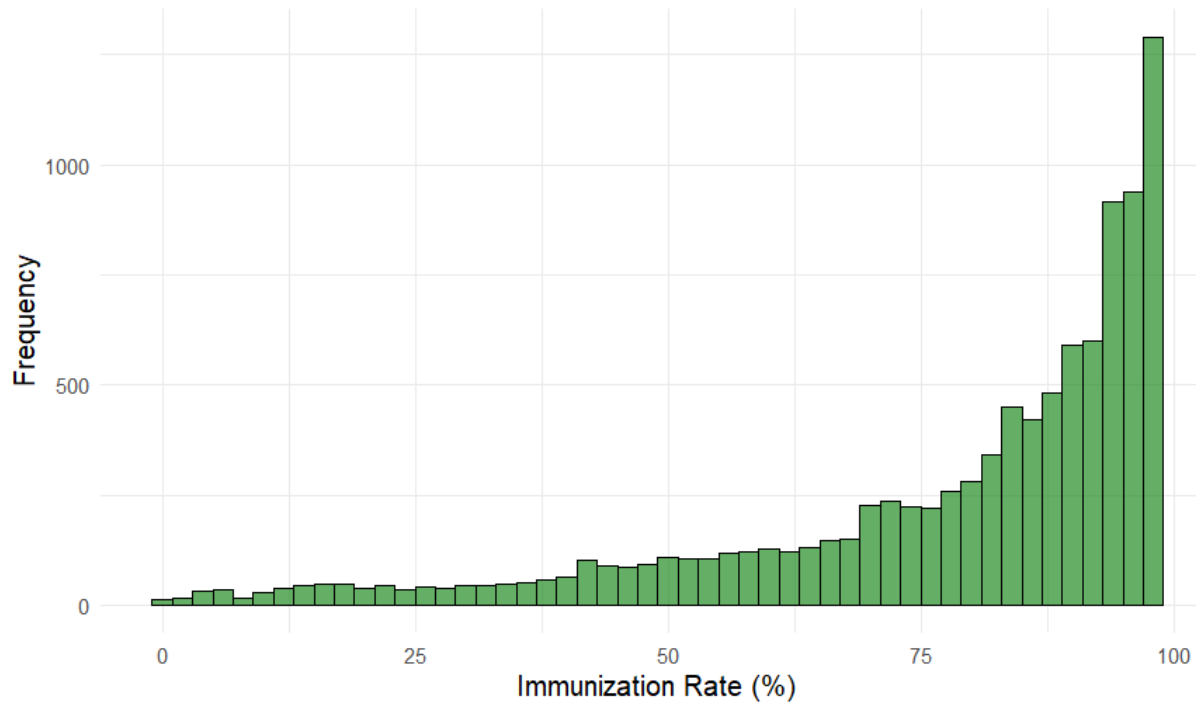


Figure 6: Immunization Rate Distribution

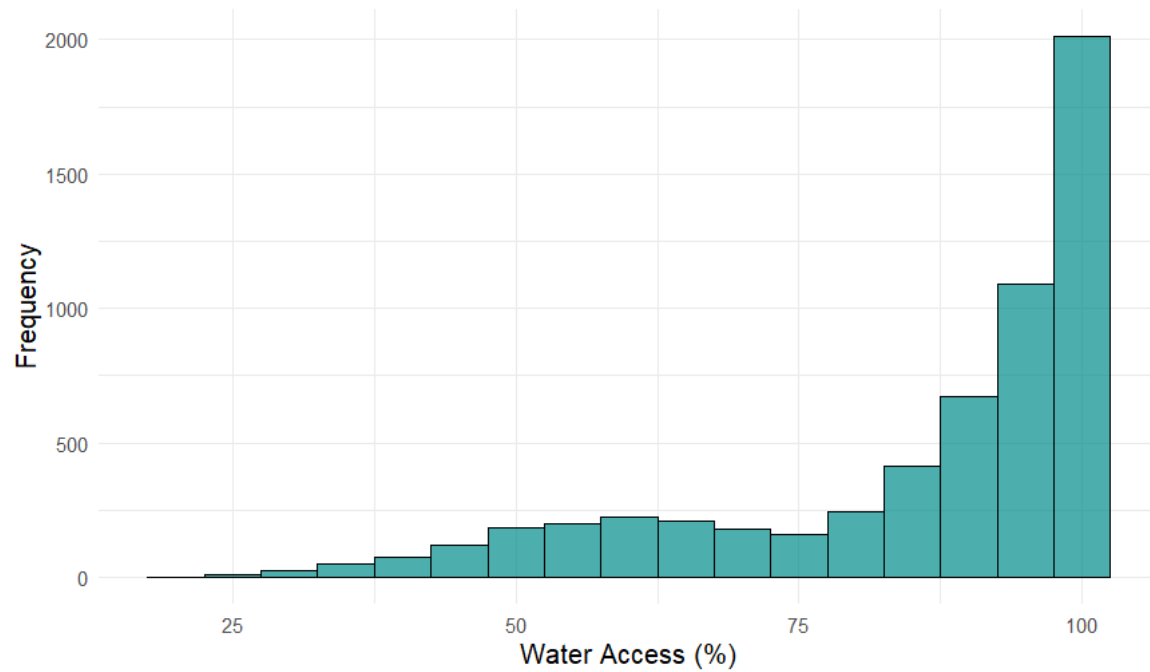


Figure 7: Water Access Distribution

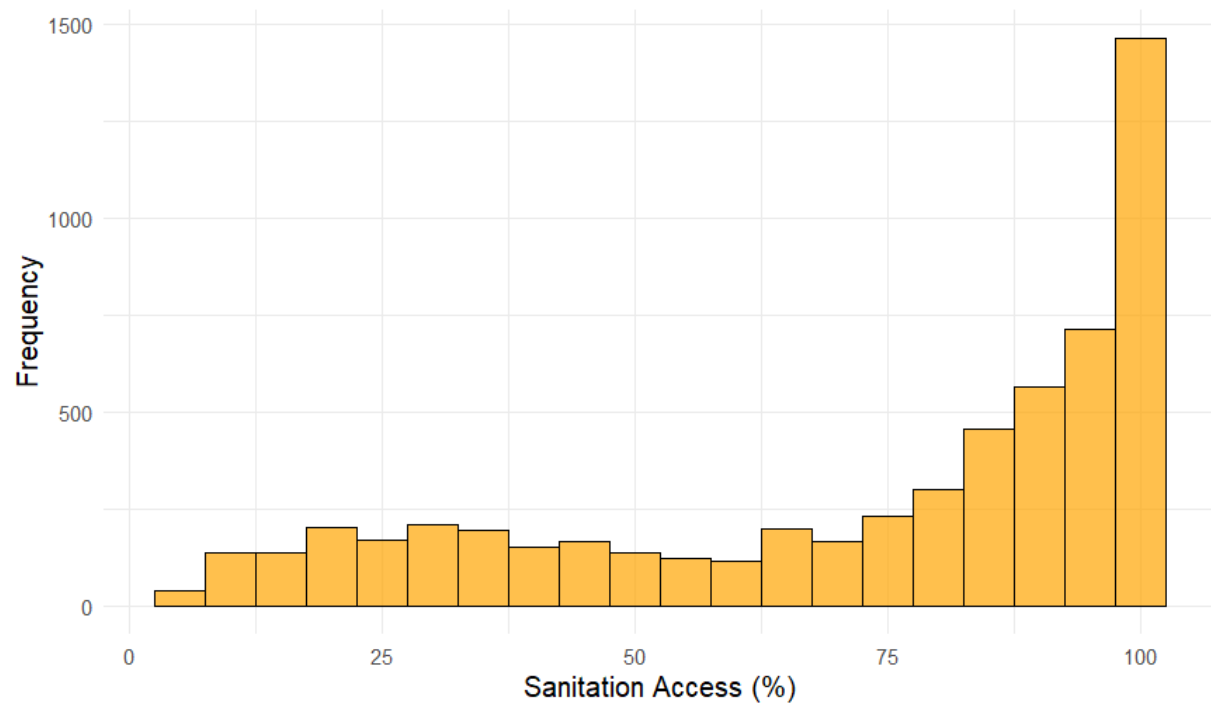


Figure 8: Sanitation Access Distribution

Scatterplots of Relationships



Figure 9: Relationship Between GDP and Life Expectancy



Figure 10: Relationship Between Unemployment and Mortality Rate

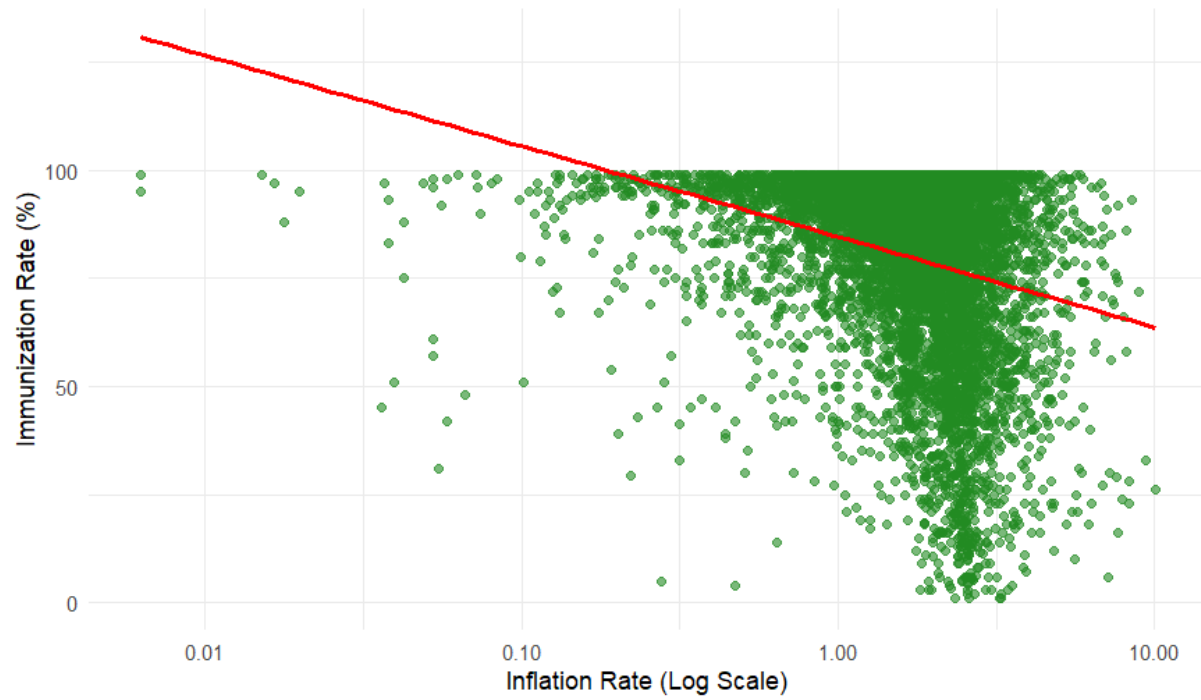


Figure 11: Relationship Between Inflation and Immunization Rate

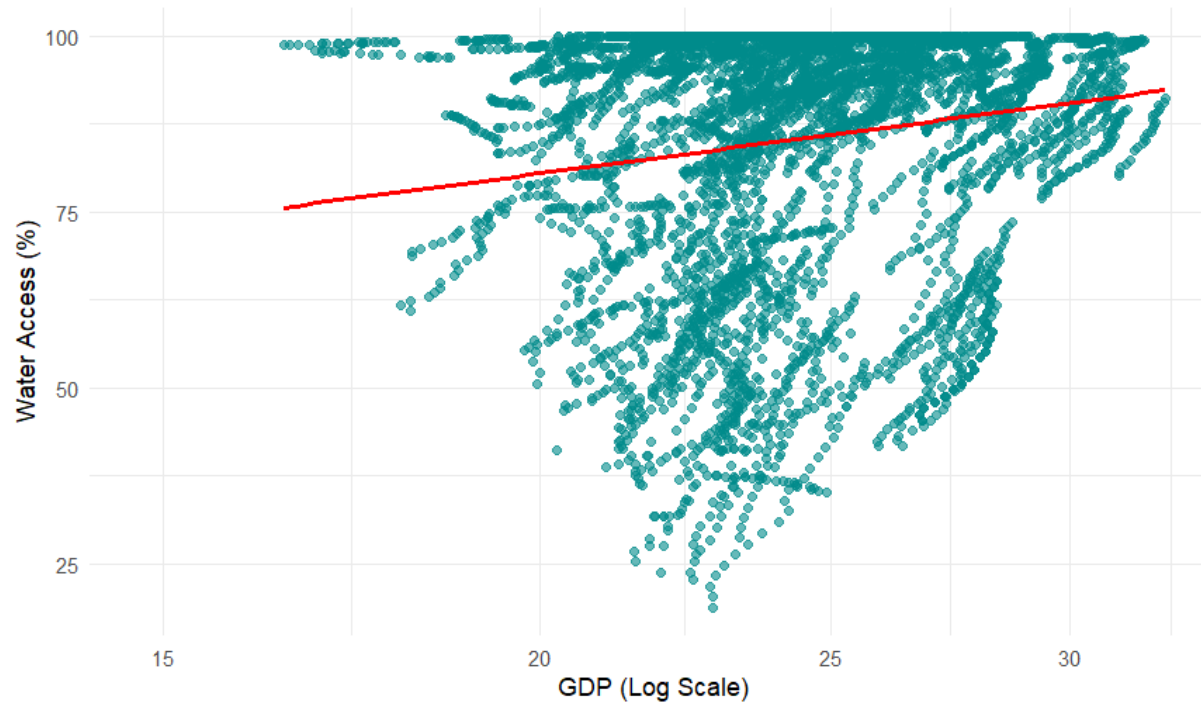


Figure 12: Relationship Between GDP and Water Access

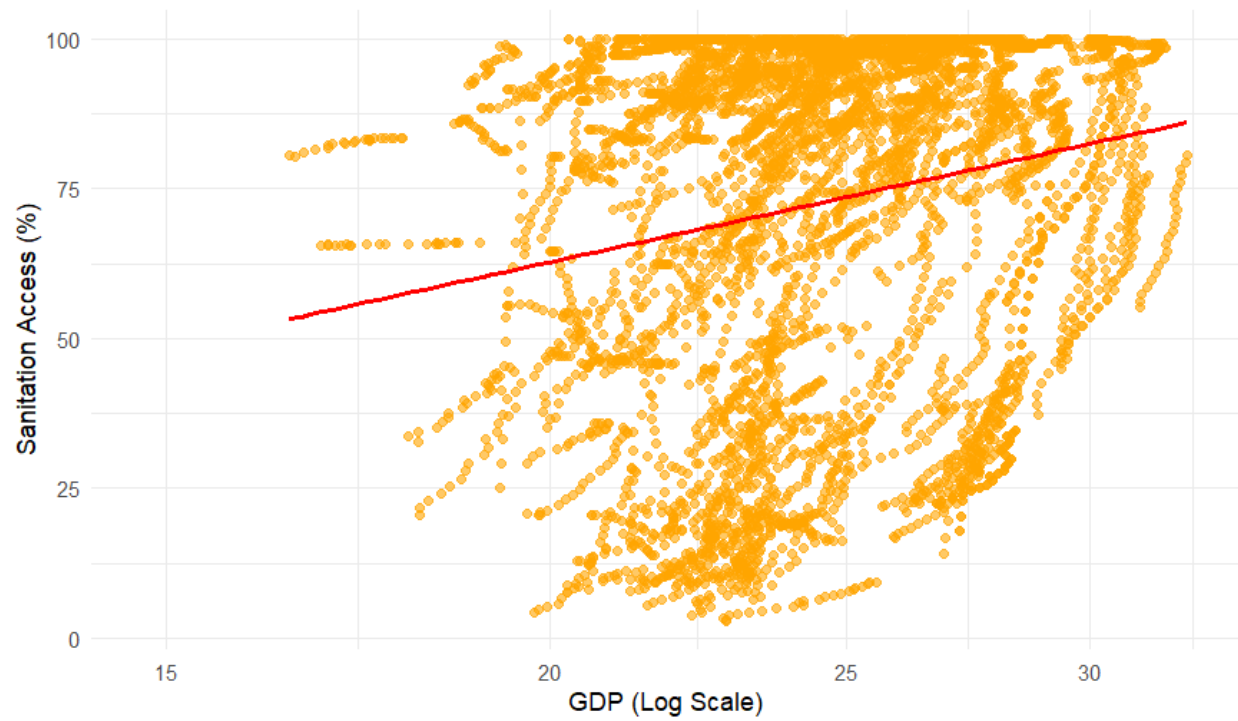


Figure 13: Relationship Between GDP and Sanitation Access



Figure 14: Relationship Between Unemployment Rate and Life Expectancy

Residual Plots of Health Indicators

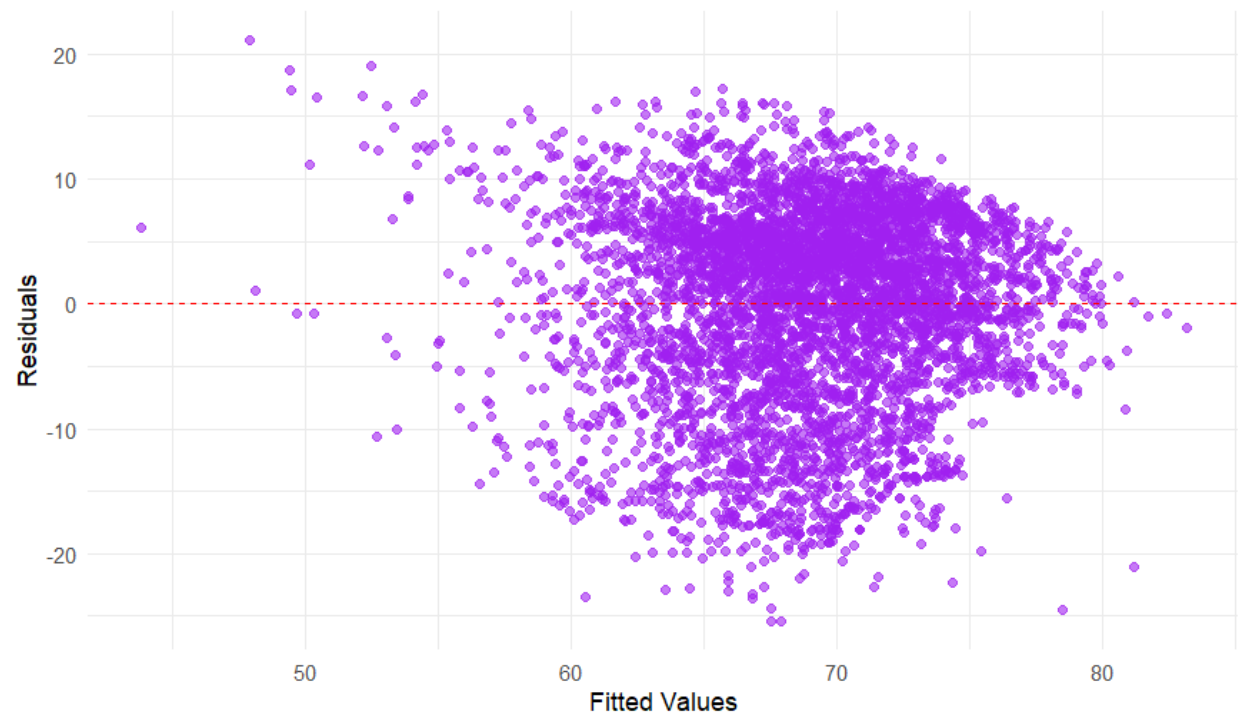


Figure 15: Residual Plot for Life Expectancy

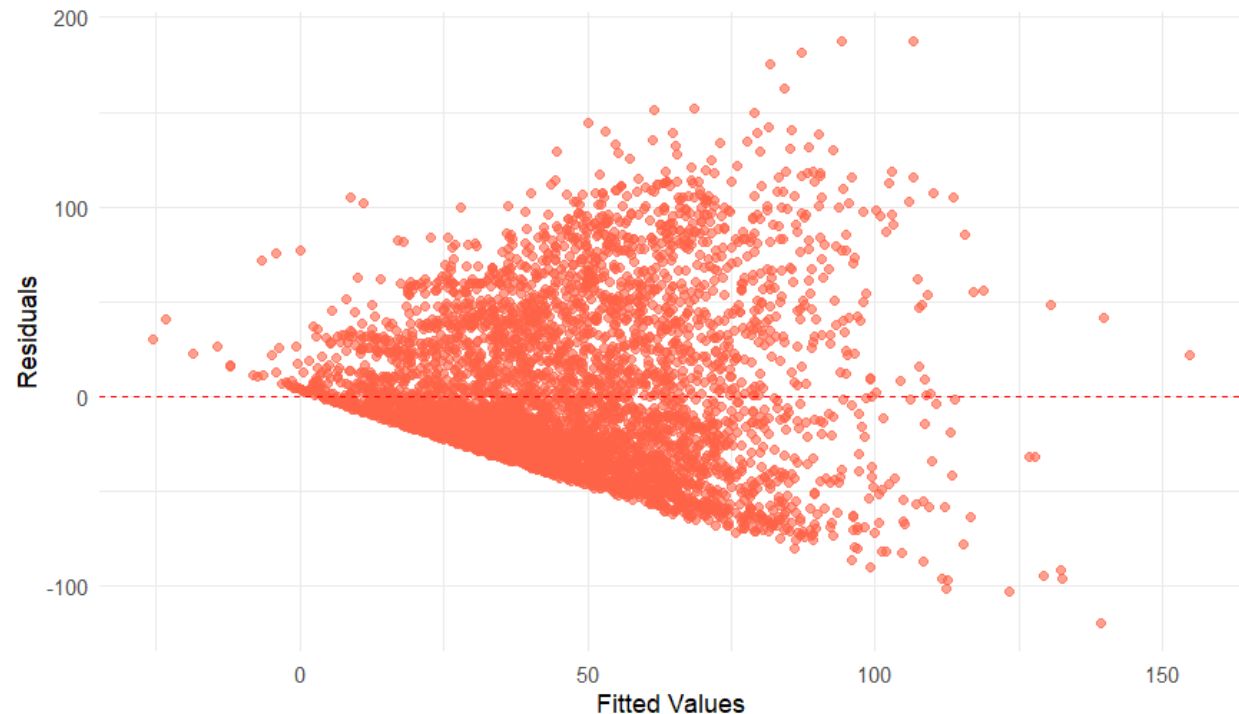


Figure 16: Residual Plot for Mortality Rate

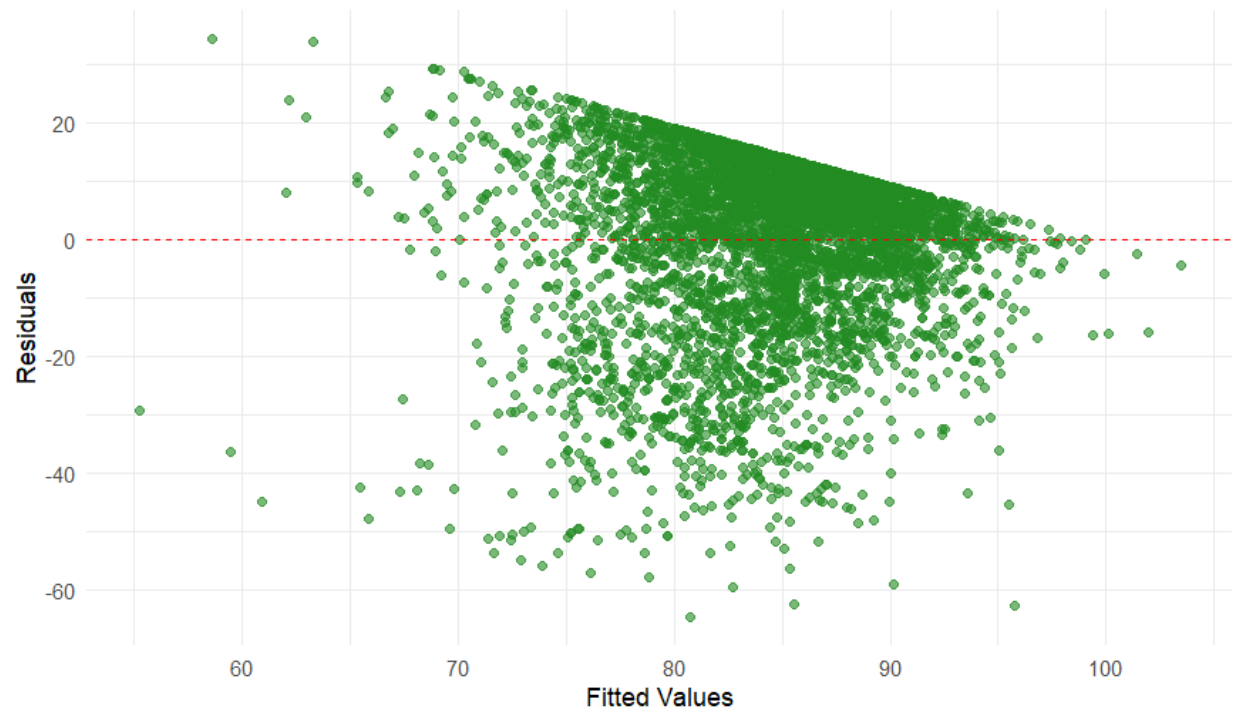


Figure 17: Residual Plot for Immunization Rate

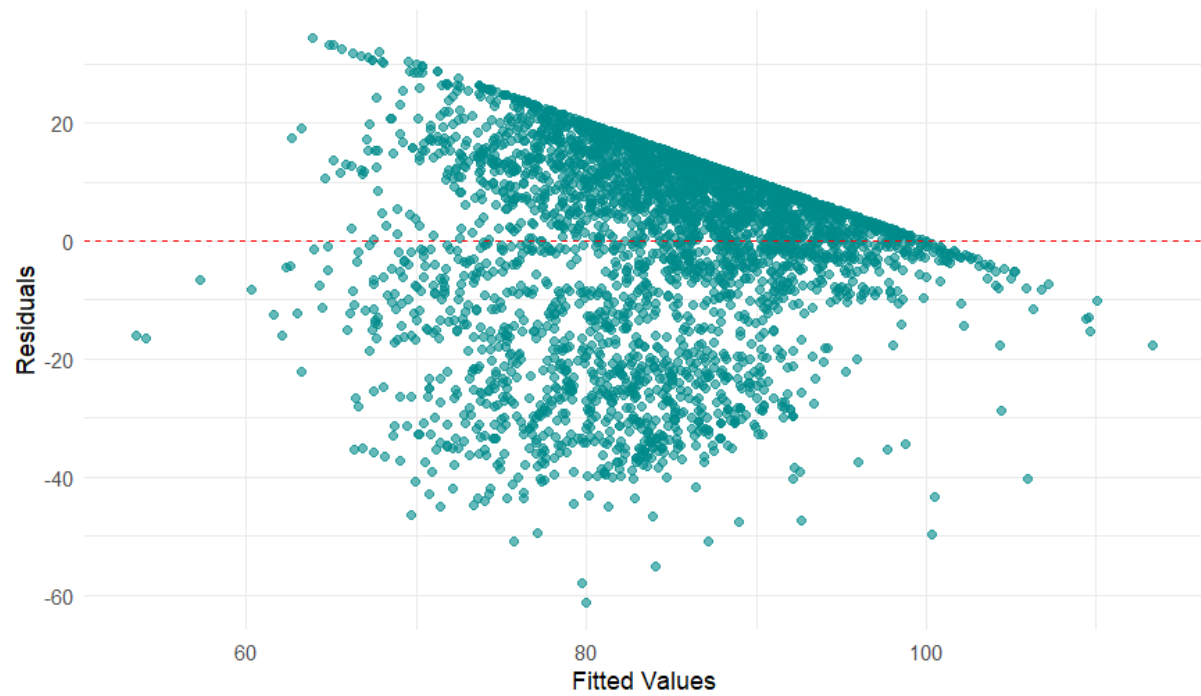


Figure 18: Residual Plot for Water Access

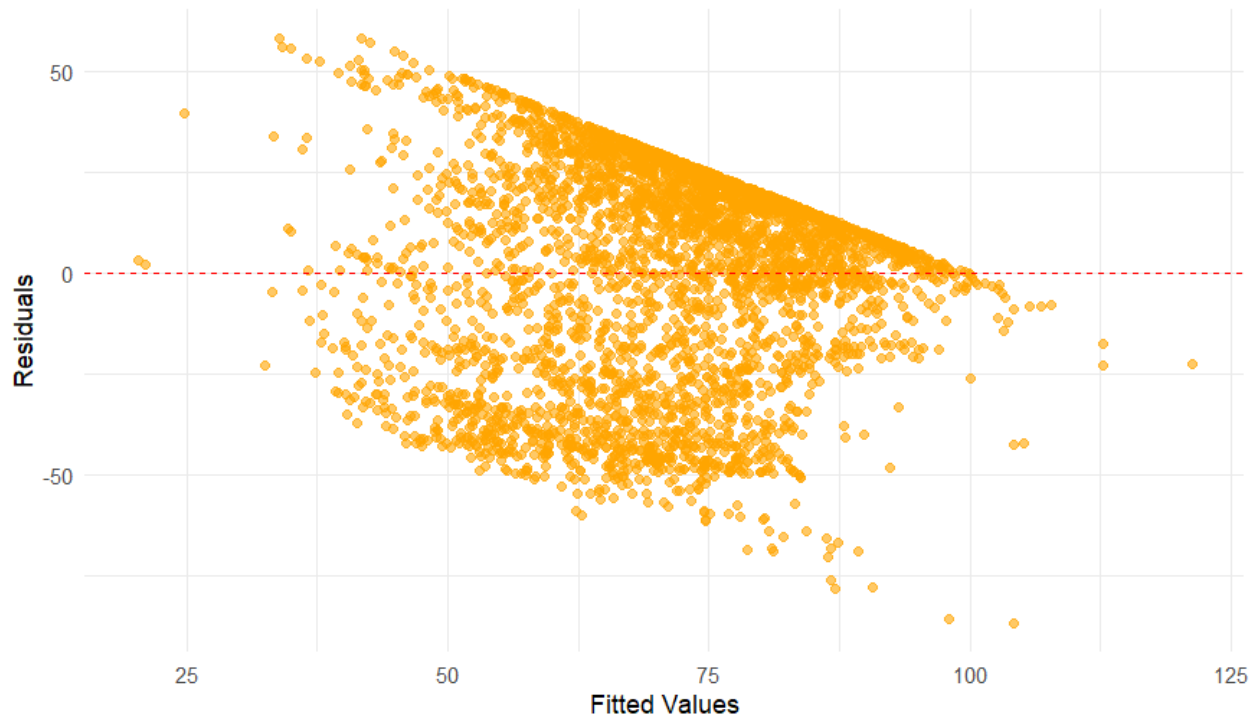


Figure 19: Residual Plot for Sanitation Access

Q-Q Plots of Health Indicators

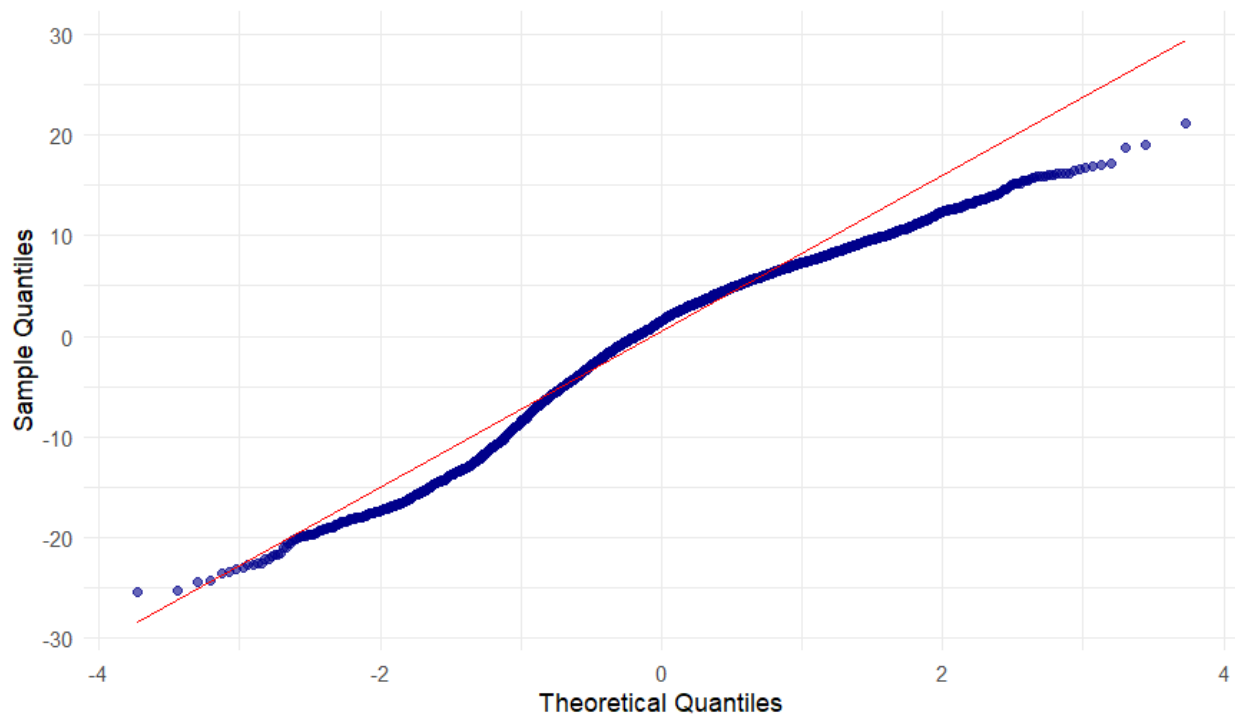


Figure 20: Q-Q Plot for Life Expectancy

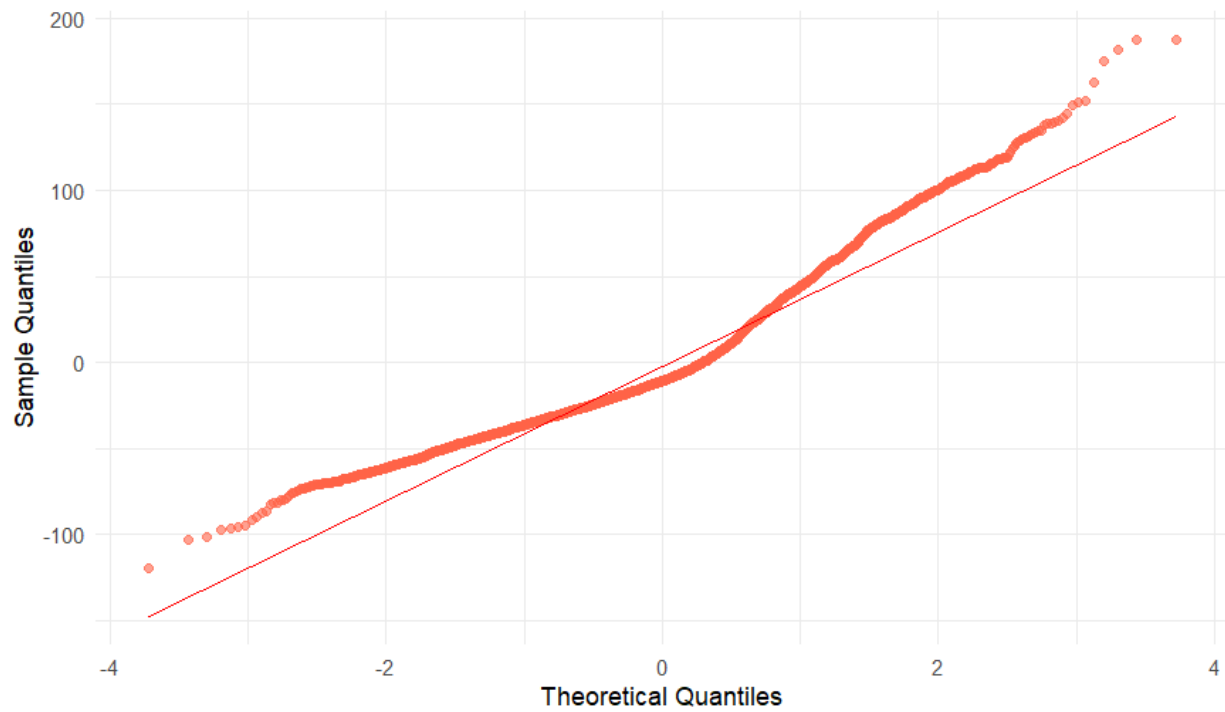


Figure 21: Q-Q Plot for Mortality Rate

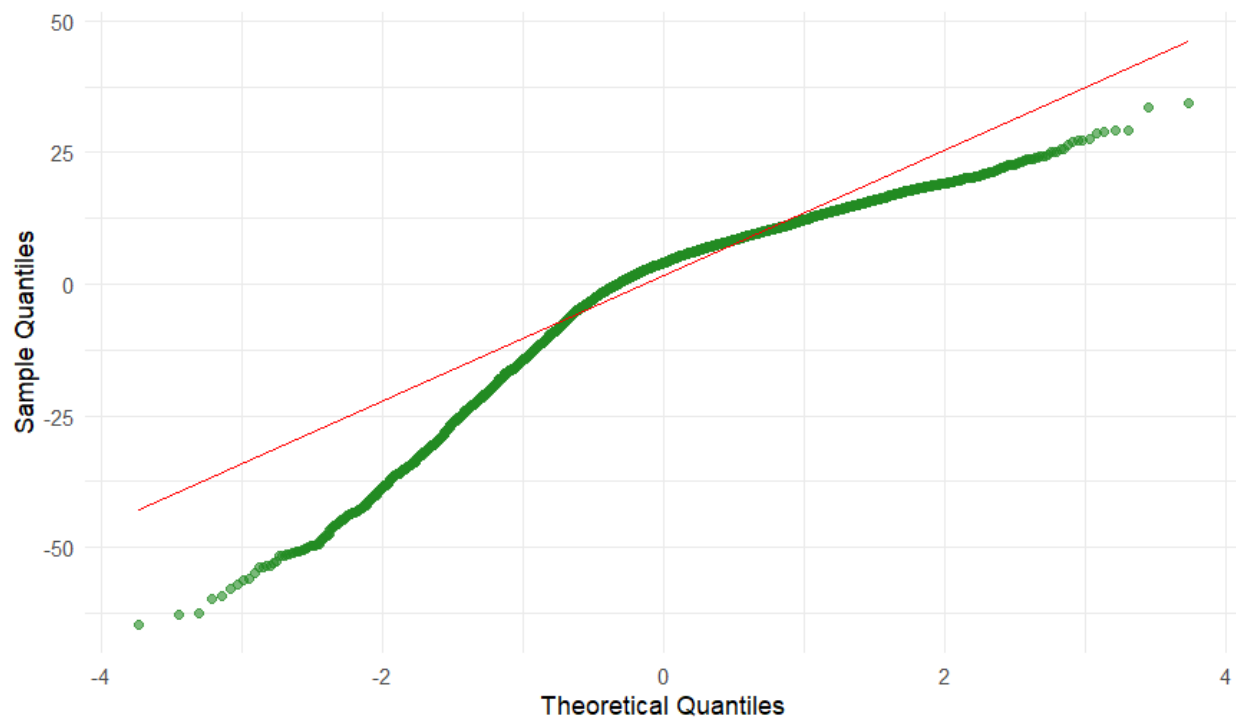


Figure 22: Q-Q Plot for Immunization Rate

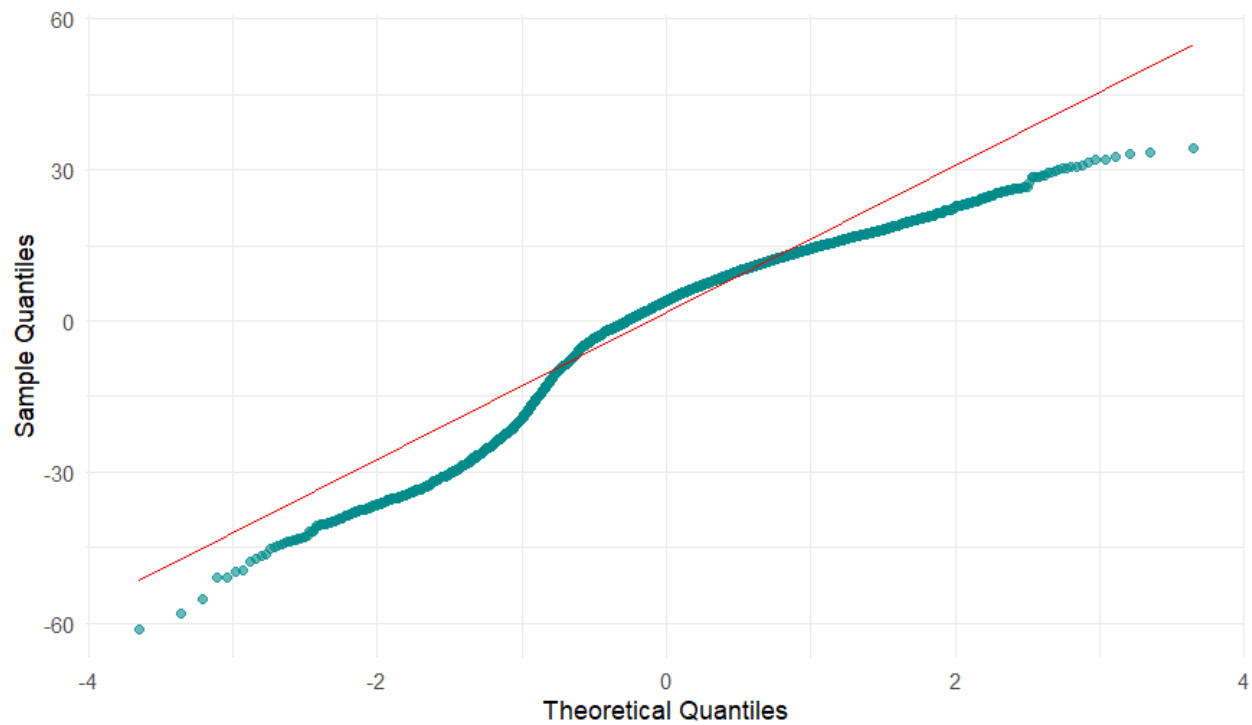


Figure 23: Q-Q Plot for Water Access

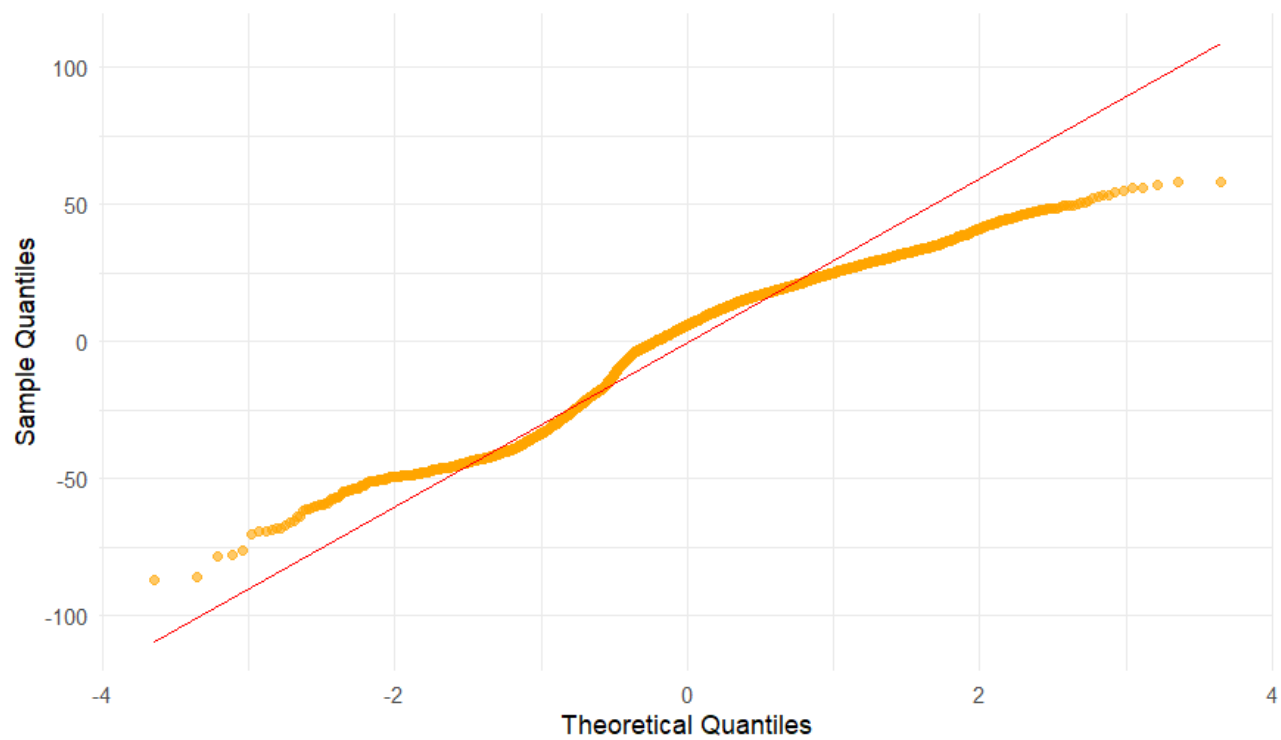


Figure 24: Q-Q Plot for Sanitation Access